

Mathematical and Computational Methods

Exercise Sheet 1: Foundations: numbers, sets and functions

This sheet accompanies *Unit 1 (Foundations: numbers, sets and functions)*. Every problem uses only the methods of Unit 1—set and interval notation, the summation and product symbols, and the elementary functions; no calculus, vectors or complex numbers are needed.

How to use this sheet. Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit1}"`) and recompile.

1 Warm-up drills

Short routine problems—at least one per skill—to build fluency.

Exercise 1.1 (★ drill — intervals & sets).

- Write $\{x \in \mathbb{R} : -2 \leq x < 5\}$ in interval notation.
- Write $(-\infty, 3]$ in set-builder notation.
- Express $\{x \in \mathbb{R} : x^2 \leq 9\}$ as a single interval.
- Compute $(-1, 4] \cap [2, 7)$.
- Compute $[0, 5] \setminus (2, 3)$.

Exercise 1.2 (★ drill — Σ and Π notation). Evaluate

$$(a) \sum_{k=1}^4 (3k-2), \quad (b) \sum_{k=1}^5 k^2, \quad (c) \prod_{k=1}^4 k, \quad (d) \prod_{k=2}^4 \left(1 - \frac{1}{k}\right).$$

Exercise 1.3 (★ drill — inequalities). Solve for real x and give the answer as an interval (or union of intervals).

$$(a) 3 - 2x \leq 7, \quad (b) x^2 - x - 6 > 0, \quad (c) |2x - 1| \leq 3, \quad (d) |x + 2| > 1.$$

Exercise 1.4 (★ drill — maximal domain). State the maximal (natural) domain of each function.

$$(a) f(x) = \frac{1}{x-2}, \quad (b) g(x) = \sqrt{x-4}, \quad (c) h(x) = \ln(x+1), \quad (d) k(x) = \frac{1}{\sqrt{3-x}}.$$

Exercise 1.5 (★ drill — composition). Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find $(f \circ g)(x)$, $(g \circ f)(x)$ and $(f \circ f)(x)$, and evaluate $(f \circ g)(3)$.

Exercise 1.6 (*★ drill — injective / surjective / bijective*). For each map $f: \mathbb{R} \rightarrow \mathbb{R}$, state whether it is injective, surjective, both (bijective) or neither.

$$(a) f(x) = 3x - 1, \quad (b) f(x) = x^2, \quad (c) f(x) = x^3, \quad (d) f(x) = e^x.$$

Exercise 1.7 (*★ drill — inverse*). Find the inverse of each function (with its domain).

$$(a) f(x) = 5x + 2 \text{ on } \mathbb{R}, \quad (b) f(x) = \sqrt{x} + 1 \text{ on } [0, \infty).$$

Exercise 1.8 (*★ drill — even / odd / periodic*). Classify each as even, odd, or neither; for (d) also state the period.

$$(a) x^4 - 3x^2, \quad (b) x^3 + x, \quad (c) x^2 + x, \quad (d) \cos(2x).$$

Exercise 1.9 (*★ drill — graph transformations*). Describe in words how the graph of $y = f(x)$ moves to give (a) $y = f(x) + 3$, (b) $y = f(x - 2)$, (c) $y = -f(x)$, (d) $y = f(2x)$. Then, starting from $y = x^2$, identify the transformations giving $y = (x + 1)^2 - 4$ and state its vertex.

Exercise 1.10 (*★ drill — hyperbolic functions*).

- (a) Evaluate $\sinh 0$, $\cosh 0$ and $\tanh 0$ from the definitions.
- (b) Is $\sinh$ even or odd? Justify from $\sinh x = \frac{1}{2}(e^x - e^{-x})$.
- (c) Without a calculator, what is the smallest possible value of $\cosh x$, and where is it attained?

Exercise 1.11 (*★ drill — the graphs of $\tan$ and $\cot$*). On separate axes, sketch $y = \tan x$ and $y = \cot x$ over $[-\pi, \pi]$. Mark every vertical asymptote and every x -intercept in that range, and state the period of each.

2 Tutorial problems

Standard-difficulty problems, anchored on the unit's worked examples, with several flavoured for the different cohorts (mathematics, data science, physics).

Problems marked ► are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 (► *★★ core — maximal domain and range*).

- (a) Find the maximal domain of $f(x) = \frac{\sqrt{x+3}}{x-1}$.
- (b) Find both the maximal domain *and* the range of $g(x) = 2 - \sqrt{4-x}$.

Exercise 2.2 (► *★★ core — inequalities, two ways*). Solve

$$(a) \frac{2x+1}{x-3} \geq 0, \quad (b) |x-1| < |x+3|.$$

Exercise 2.3 (*★★ core — Σ manipulation (maths)*).

- (a) Using $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, show that $\sum_{k=1}^n (3k+2) = \frac{3n^2+7n}{2}$, and evaluate it at $n = 10$.
- (b) Prove that $\sum_{k=1}^n (2k-1) = n^2$ by splitting the sum.

Exercise 2.4 (► *★★ core — composition and decomposition*). Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$.

- (a) Find $(f \circ g)(x)$ and its domain, and $(g \circ f)(x)$ and its domain.
- (b) Express $h(x) = (2x + 1)^5$ and $p(x) = \frac{1}{\sqrt{x^2 + 1}}$ each as a composition of simpler functions.

Exercise 2.5 (► **core — the same rule, different verdicts). Consider the rule $f(x) = x^2$.

- (a) As a map $\mathbb{R} \rightarrow \mathbb{R}$, show it is neither injective nor surjective.
- (b) Show that as a map $[0, \infty) \rightarrow [0, \infty)$ it is bijective, and write down its inverse.

Exercise 2.6 (**core — an inverse via $y = x^3$).

- (a) Explain why $f(x) = x^3$ is bijective on $\mathbb{R}$, and find f^{-1} .
- (b) Find the inverse of $g(x) = (x - 1)^3 + 2$ and verify that $g(g^{-1}(x)) = x$.

Exercise 2.7 (► **core — graph transformations).

- (a) Obtain $g(x) = (x - 2)^2 + 1$ from $y = x^2$ by a sequence of transformations; state the vertex and sketch both curves.
- (b) Obtain $h(x) = 3 - \sqrt{x + 2}$ from $y = \sqrt{x}$. State the domain, the transformations used, and whether h is increasing or decreasing.

Exercise 2.8 (**core — even and odd parts (maths)).

- (a) Prove that the product of two odd functions is even.
- (b) Show that any function f on a domain symmetric about 0 can be written $f = E + O$ with $E(x) = \frac{1}{2}(f(x) + f(-x))$ even and $O(x) = \frac{1}{2}(f(x) - f(-x))$ odd. Apply this to $f(x) = e^x$.

Exercise 2.9 (**core — reading the piecewise ReLU). Recall $\text{ReLU}(x) = \max(0, x)$.

- (a) Write ReLU as a piecewise formula, and evaluate it at $x = -3, 0, 2$.
- (b) Show that $\text{ReLU}(x) = \frac{1}{2}(x + |x|)$.
- (c) Hence express $|x|$ in terms of $\text{ReLU}(x)$ and $\text{ReLU}(-x)$.

Exercise 2.10 (**core — exponentials and logarithms). Solve for x (exact values).

- (a) $2^x = 10$, (b) $3e^{2x} = 12$, (c) $\ln 8 - \ln 2 + \ln \frac{1}{2}$ (simplify to a single log).

Exercise 2.11 (► **core — trigonometric equations). Solve on $[0, 2\pi)$, giving exact answers in radians.

- (a) $\cos x = \frac{1}{2}$, (b) $\tan x = 1$, (c) $2 \sin x - 1 = 0$.

Exercise 2.12 (► **core — the sigmoid as a probability (data science)). The logistic (sigmoid) function is $\sigma(x) = \frac{1}{1 + e^{-x}}$.

- (a) Compute $\sigma(0)$ and explain why $0 < \sigma(x) < 1$ for every real x .
- (b) Verify the symmetry $\sigma(-x) = 1 - \sigma(x)$.
- (c) In logistic regression a real-valued score s is turned into a class probability $\sigma(s)$. If $s = \ln 4$, find the probability. What score gives probability $\frac{1}{2}$?

Exercise 2.13 (► **core — a potential barrier from steps (physics)). The rectangular potential barrier of height $V_0 > 0$ and width $2a$ is $V(x) = V_0$ for $-a \leq x \leq a$ and $V(x) = 0$ otherwise. Throughout, H is the Heaviside step, with the convention $H(0) = 1$ (so $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$).

- (a) What is the range of V ? Where is V discontinuous?
- (b) Show that $V(x) = V_0[H(x+a) - H(x-a)]$, and check the value at $x = 0$ and at $x = 2a$.

Exercise 2.14 (► ★★ core — the inverse-trig function arctan). arctan is the inverse of tan restricted to the branch $(-\frac{\pi}{2}, \frac{\pi}{2})$.

- (a) State the maximal domain and the range of arctan.
- (b) Explain why the lines $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes, and why arctan is an odd function.
- (c) Evaluate arctan 0 and arctan 1, and sketch $y = \arctan x$.

Exercise 2.15 (★★ core — sketching the sigmoid (data science)). For the logistic (sigmoid) function $\sigma(x) = \frac{1}{1+e^{-x}}$, produce a careful hand sketch. On your graph mark: the value $\sigma(0)$, the two horizontal asymptotes, the direction of monotonicity, and the centre of symmetry. Justify each feature in one line.

3 Further practice (home)

A larger bank for independent study, rising from routine to harder, with conceptual checks (true/false and spot-the-error) before a closing block of ★★ challenge challenge / extension problems.

Exercise 3.1 (★★ core — notation round-up).

- (a) Write $\{x \in \mathbb{R} : 1 < 2x - 3 \leq 7\}$ as an interval.
- (b) Write $[-2, 0) \cup (0, 2]$ in set-builder notation in one line.
- (c) Evaluate $\sum_{k=0}^4 2^k$ and $\prod_{k=1}^n \frac{k+1}{k}$.

Exercise 3.2 (★★ core — a rational inequality). Solve $\frac{x}{x+1} < 2$.

Exercise 3.3 (★★ core — maximal domains, harder). State the maximal domain of each.

$$(a) f(x) = \ln(x^2 - 4), \quad (b) g(x) = \frac{\sqrt{x^2 - 1}}{x - 3}, \quad (c) h(x) = \arcsin(x - 1).$$

Exercise 3.4 (★★ core — compositions with reciprocals). Let $f(x) = \frac{1}{x}$ and $g(x) = x + 1$.

- (a) Find $f \circ g$, $g \circ f$ and $f \circ f$, each with its domain.
- (b) Decompose $F(x) = \sqrt{\ln x}$ into simpler functions and state its domain.

Exercise 3.5 (★★ core — inverses, including a Möbius map). Find the inverse (with domain) of

$$(a) f(x) = 2 + \sqrt{x-1}, \quad (b) g(x) = \frac{3x+1}{x-2}.$$

Exercise 3.6 (★★ core — classify the maps). Decide injective / surjective / bijective.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 - x$.
- (b) $f: (-\infty, 0] \rightarrow [0, \infty), f(x) = x^2$.

Exercise 3.7 (**core — *symmetry zoo*). Classify each as even, odd or neither; note any obvious period.

$$x^5 \sin x, \quad x|x|, \quad e^{-x^2}, \quad x + \cos x, \quad |\sin x|.$$

Exercise 3.8 (**core — *transforming standard graphs*).

- (a) From $y = \frac{1}{x}$, describe the graph of $y = \frac{1}{x-2} + 1$, giving both asymptotes.
 (b) From $y = \sin x$, describe the graph of $y = 2 \sin(x) - 1$, giving its range.

Exercise 3.9 (**core — *log and exponential equations*). Solve for x .

$$(a) \log_2 x + \log_2(x-2) = 3, \quad (b) e^{2x} - 5e^x + 6 = 0.$$

Exercise 3.10 (**core — *more trigonometric equations*). Solve on $[0, 2\pi)$.

$$(a) 2 \cos^2 x - 1 = 0, \quad (b) \sin x = \cos x.$$

Exercise 3.11 (**core — *the logit (data science)*). For the sigmoid $\sigma(x) = \frac{1}{1+e^{-x}}$ (range $(0, 1)$):

- (a) Show that its inverse is the *logit* $\sigma^{-1}(p) = \ln \frac{p}{1-p}$, $0 < p < 1$.
 (b) A classifier outputs probability $p = 0.8$. What score s produced it?

Exercise 3.12 (**core — *a pulse from steps (physics)*). Using the Heaviside step H , write a single formula for the signal that is 0 for $t < 0$, equal to 5 for $0 \leq t < 2$, and 0 for $t \geq 2$. Check it at $t = 1$ and $t = 3$.

Exercise 3.13 (**core — *hyperbolic identities*). Work directly from $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$.

- (a) Prove $\cosh^2 x - \sinh^2 x = 1$.
 (b) Prove $\sinh(2x) = 2 \sinh x \cosh x$.

Exercise 3.14 (**core — *finding ranges*). Find the range of each function.

$$(a) f(x) = \frac{x^2}{x^2 + 1}, \quad (b) g(x) = 2 \sin x + 1.$$

Exercise 3.15 (**core — *true or false?*). State whether each is true or false, with a one-line reason.

- (a) $\infty \in \mathbb{R}$.
 (b) Every injective function $\mathbb{R} \rightarrow \mathbb{R}$ is surjective.
 (c) $f \circ g = g \circ f$ for all functions f, g .
 (d) The maximal domain of $\ln(x^2)$ is all of $\mathbb{R}$.
 (e) Every periodic function is bounded.

Exercise 3.16 (**core — *spot the error*). Each “solution” below contains a mistake. Find and correct it.

- (a) “ $-2x > 6$, so $x > -3$.”
 (b) “The maximal domain of $\sqrt{x-2}$ is $x \leq 2$.”
 (c) “ $|x| = -2$ has the solutions $x = \pm 2$.”

Exercise 3.17 (***challenge — irrationality of $\sqrt{3}$). Prove that $\sqrt{3}$ is irrational, imitating the proof for $\sqrt{2}$ in the notes.

Exercise 3.18 (***challenge — the Gauss sum). Prove $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ by writing the sum forwards and backwards and adding, with no appeal to induction.

Exercise 3.19 (***challenge — monotonic $\Rightarrow$ injective, but not conversely).

- (a) Prove that a strictly increasing function is injective.
- (b) Give an injective function on $[0, 2]$ that is *not* monotonic.

Exercise 3.20 (***challenge — sigmoid and $\tanh$). Show that $\sigma(x) = \frac{1 + \tanh(x/2)}{2}$.